

① Work : Transfer of energy due to displacement by a force.

→ +ve OR -ve

↓
Supply of
Energy

↓
Gain of
Energy

PG
Physics Galaxy
Videos

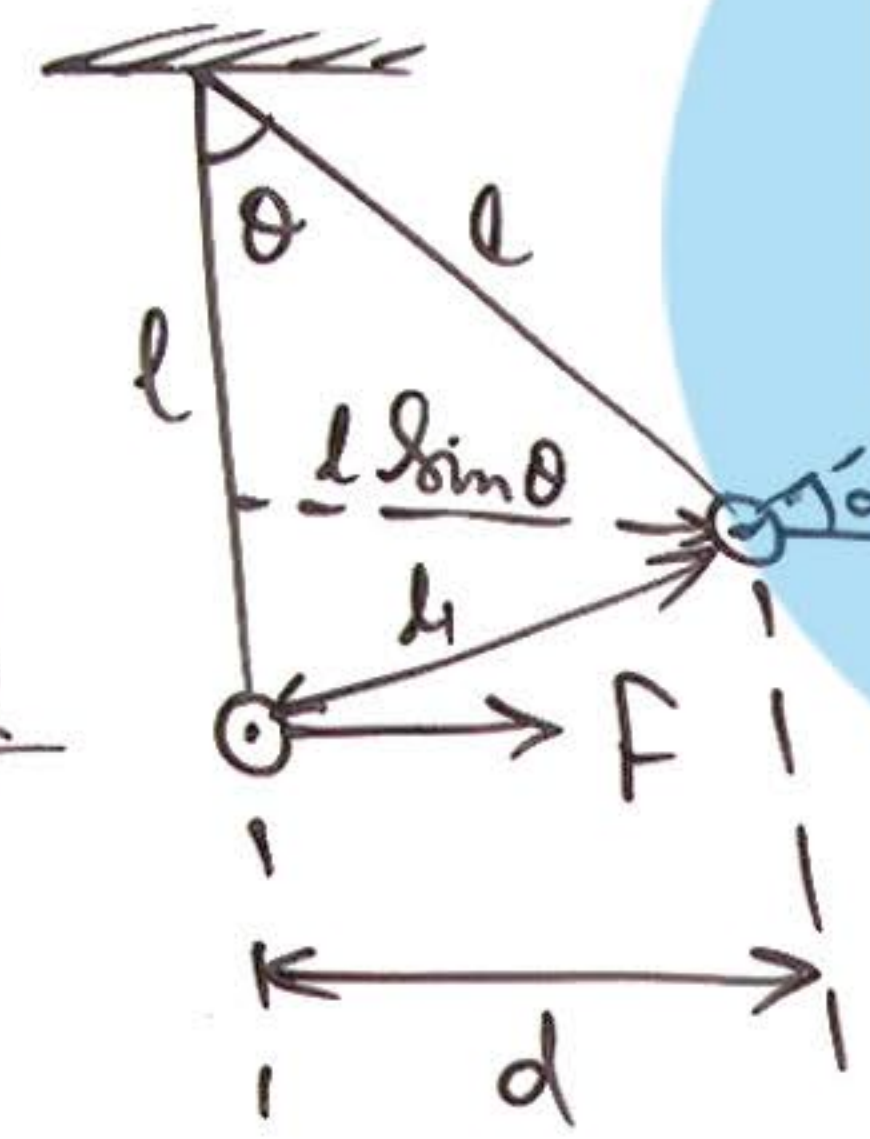
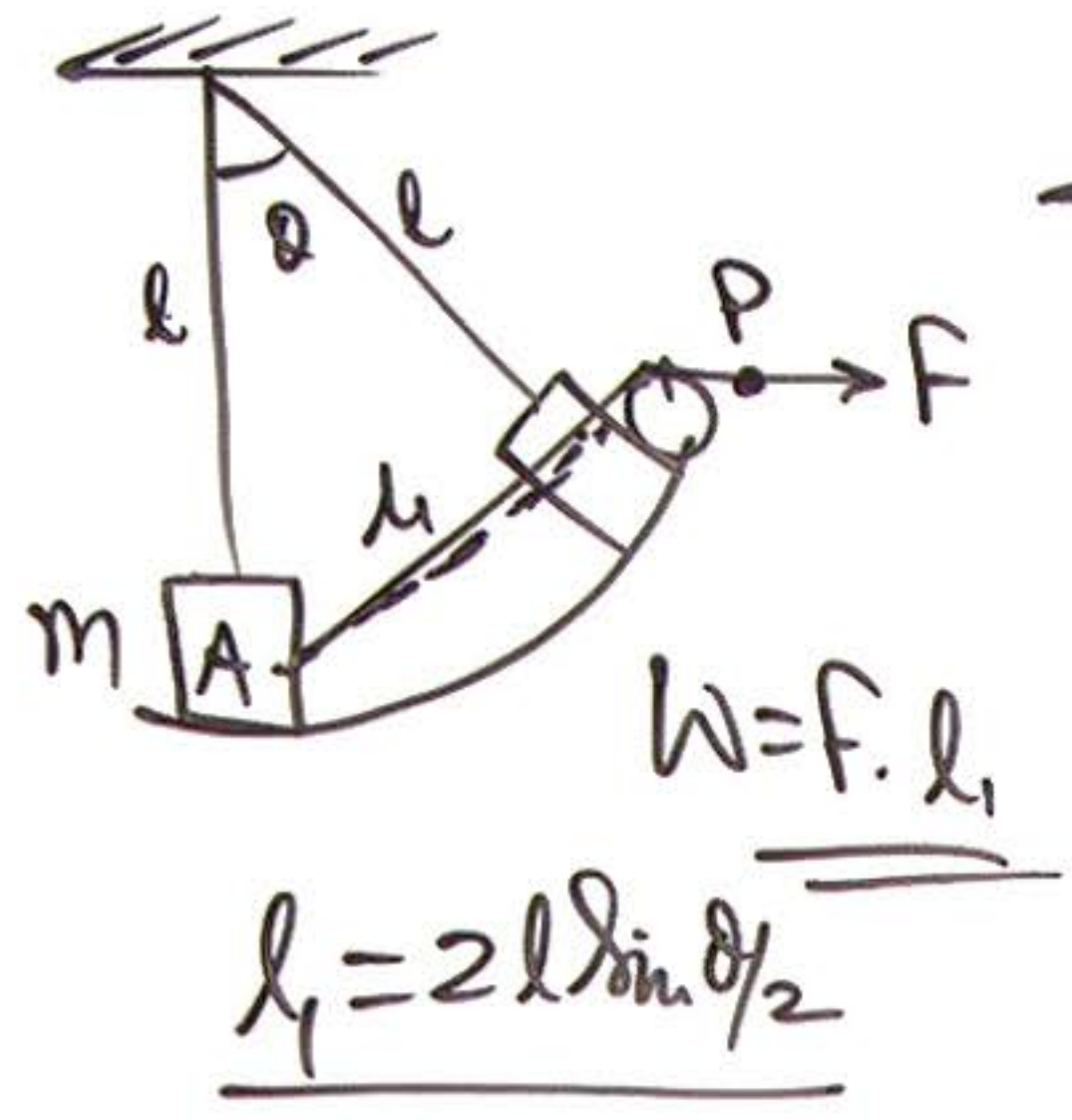
② Calculation of work:

→ Constant force. →

displacement in the dir of force.

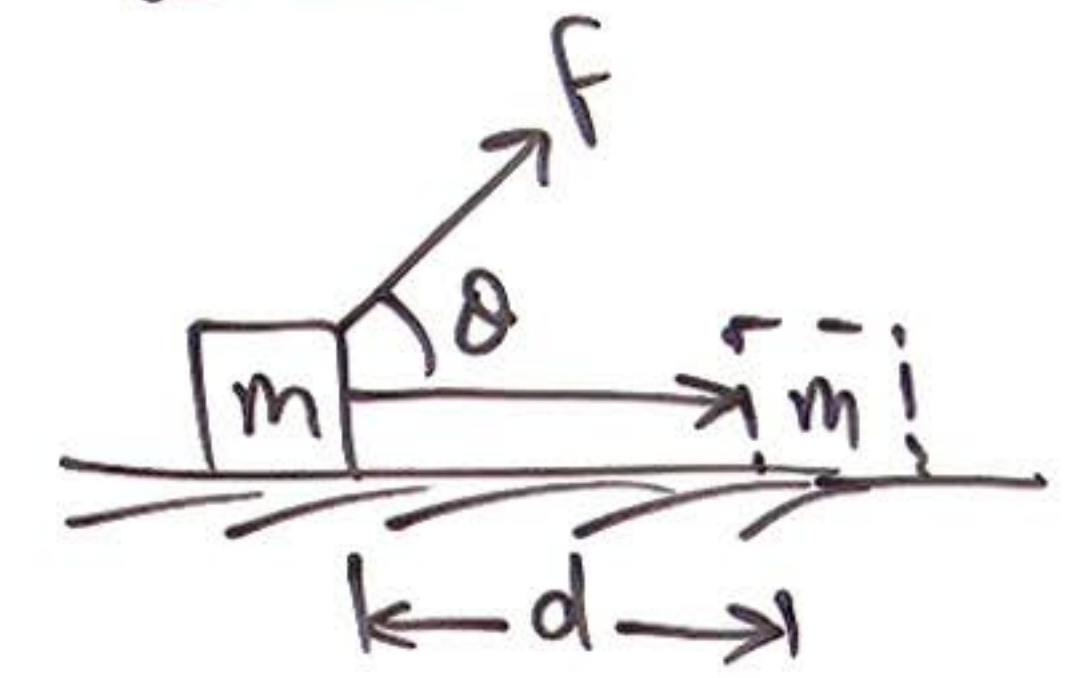
$$W = F \cdot d$$

$$W = \vec{F} \cdot \vec{d} = \underline{\underline{F d \cos \theta}}$$

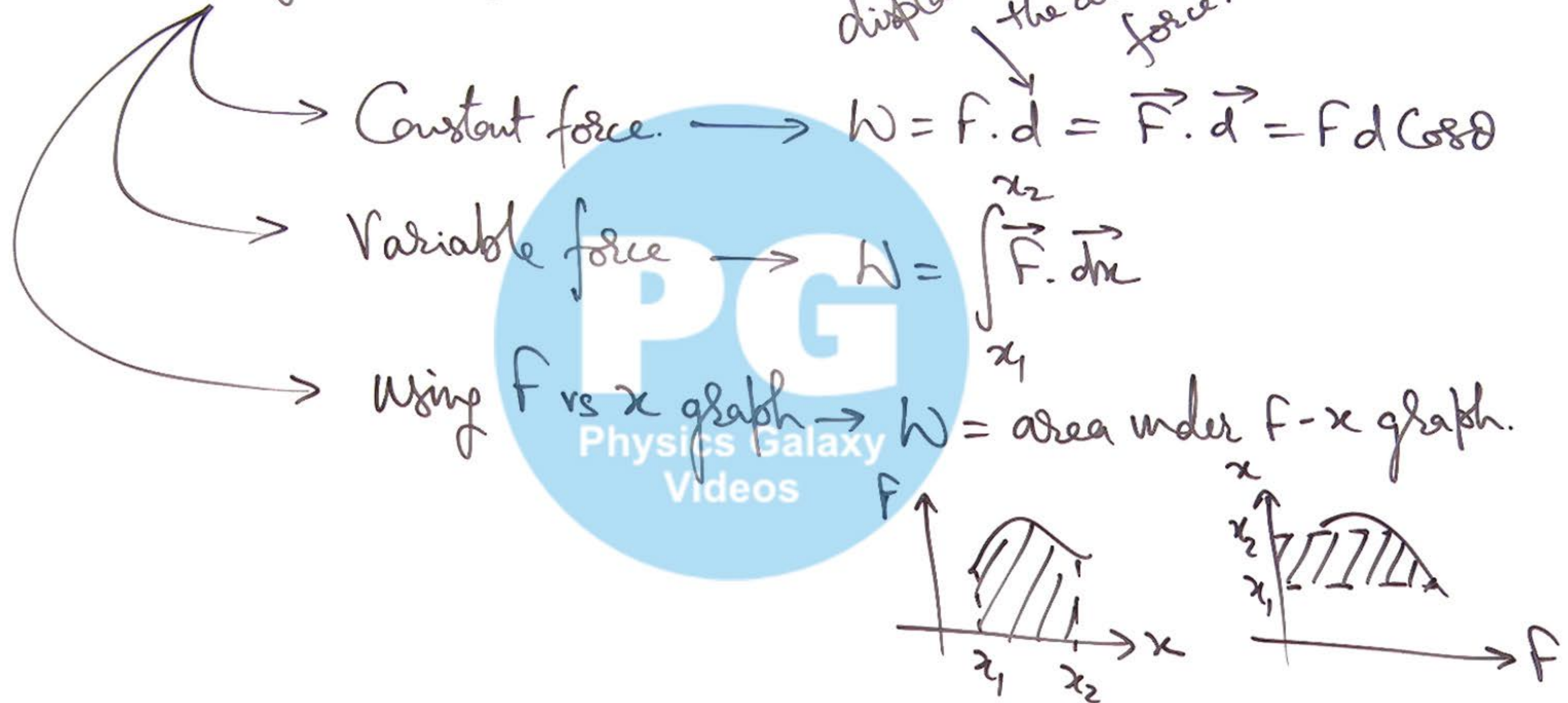


$$W = F \cdot d = F \cdot l \sin \theta$$

$$= F \cdot l_1 \cos \alpha$$



② Calculation of work:



③ Energy & Its Types:

$\xrightarrow{\text{Mech. Energy}}$
 $\rightarrow KE$ - always +ve ($\frac{1}{2}mv^2$)
 $\rightarrow PE^*$ - It is a stored form of energy. $\rightarrow$ It is associated with some force field.

④ Forces & Types in

respect of energy Conservation
 PG
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 Videos

$\rightarrow$ $\star$ Conservative force $\rightarrow$ Cause change in PE of system

$\rightarrow$ Non Conservative force. $\rightarrow$ Cause dissipation of energy in system.
 $\downarrow$
 loss to surrounding

⑤ Force & PE Relationship:

$$\vec{F} = - \overset{\substack{\text{gradient} \\ \downarrow}}{\nabla} U \quad \left[\nabla \equiv \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right]$$

$$\checkmark \quad \vec{F} = - \left[\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k} \right]$$

In One dimensional variation, we use

$$\boxed{\vec{F} = - \frac{dU}{dx} \hat{i}}$$

$$\boxed{\vec{F} = - \frac{dU}{dz} \hat{k}}$$

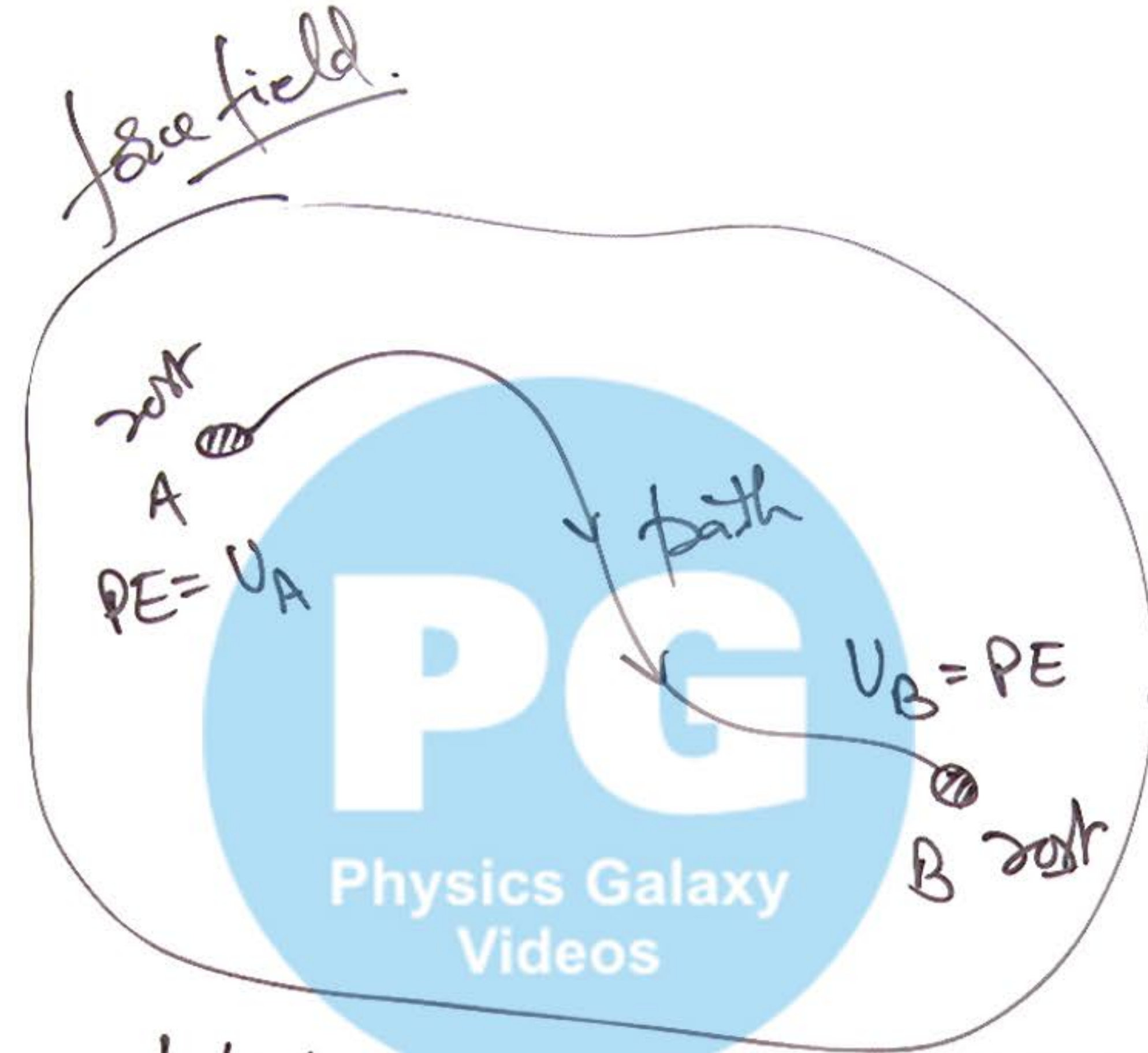
Ex: $U = (x^2 y + x z^3) \text{ J}$
find Force at (1, 1, 2)

$$\vec{F} = - \left[(2xy + z^3) \hat{i} + x^2 \hat{j} + 3xz^2 \hat{k} \right]_{(1,1,2)}$$

$$\underline{\vec{F} = -10\hat{i} - \hat{j} - 12\hat{k} \text{ N}}$$

⑥ Work & PE:

↓
(transfer
of energy)



W/D by external forces

Work done by
System/force field/
internal force.

$$W = U_A - U_B$$

(i) (f)

$$W = U_B - U_A$$

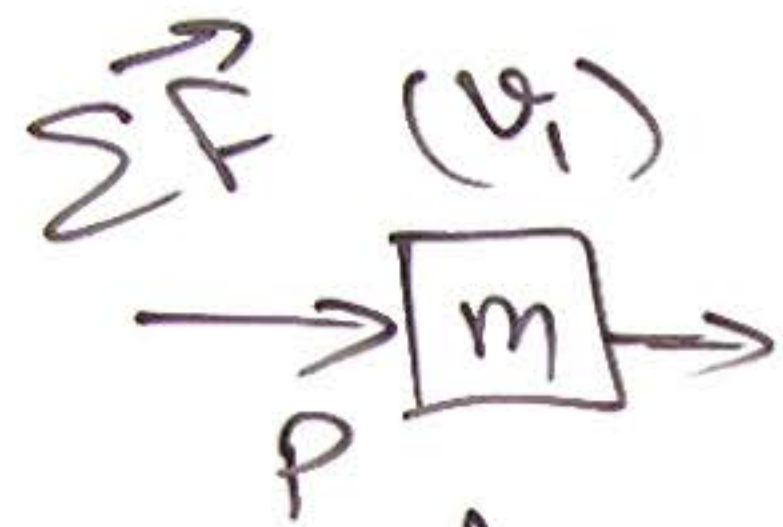
(f) (i)

$$KE_i = KE_f = 0$$

⑦ Work-Energy Theorem: $\rightarrow$ 2nd Exp for W/E theorem.

(for rigid bodies)

$$KE_i + \text{W/D by } \underset{\substack{\downarrow \\ \text{accelerating force}}}{\text{Supporting force}} - \text{W/D by } \overset{\substack{\leftarrow \\ \text{retarding force}}}{\text{opposing forces}} = KE_f.$$



A(i)

$$KE_i = \frac{1}{2}mv_i^2$$



B(f)

$$KE_f = \frac{1}{2}mv_f^2$$

if $KE_f > KE_i \Rightarrow$ W/D by body P $\rightarrow$ -ve.

if $KE_f < KE_i \Rightarrow$ W/D by body P $\rightarrow$ +ve

ext W/D on P.
body
+ve
-ve

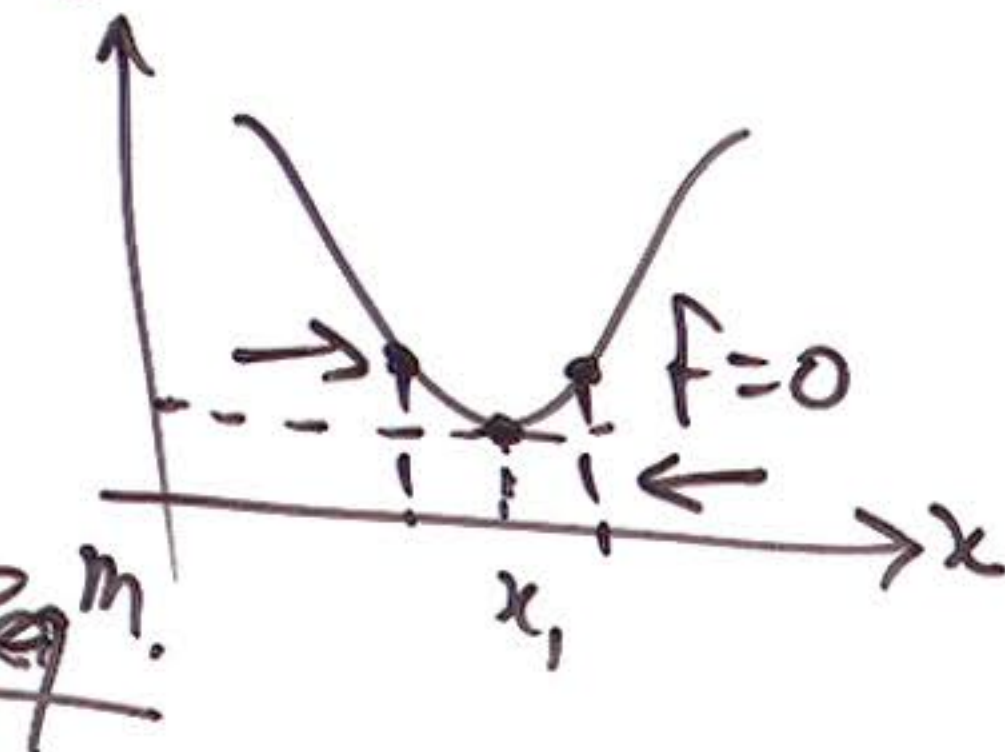
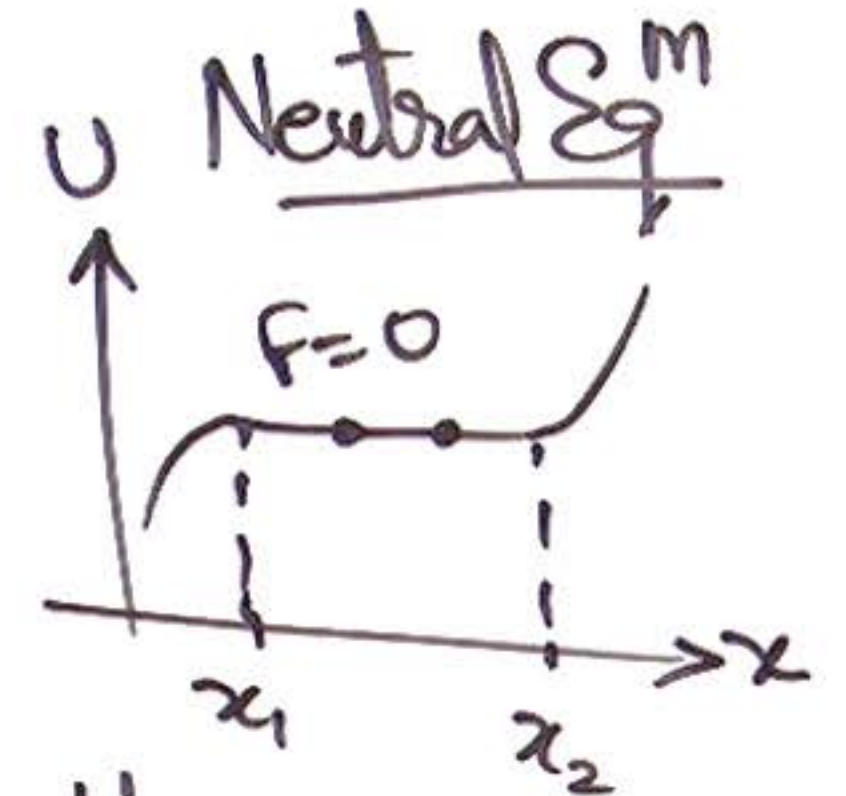
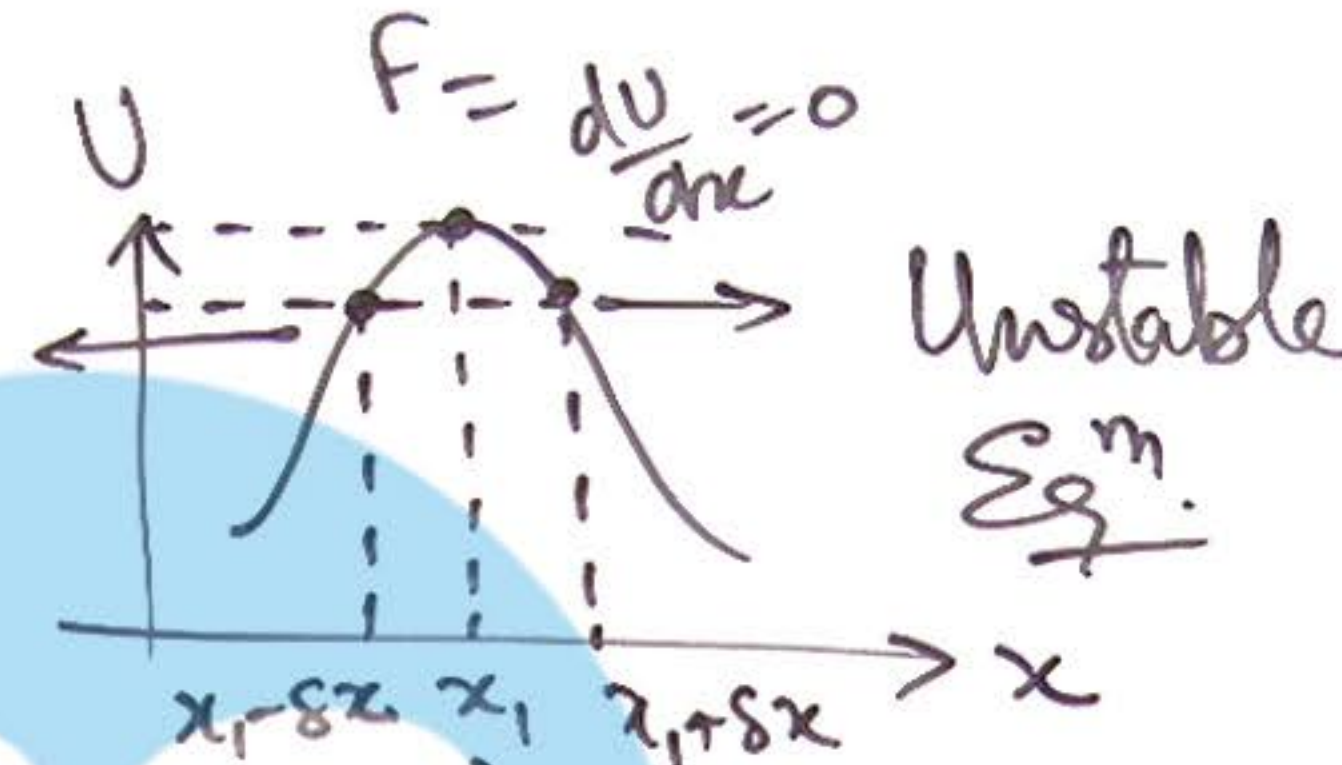
⑧ Concept of Equilibrium & PE:

Conservative forces

$$\vec{F} = -\frac{dU}{dx} \hat{i}$$

Under conservative forces

if body is in equilibrium $\Rightarrow \frac{dU}{dx} = 0$



Oscillation $\leftarrow$ Stable Eg^m .

$U = \text{Constant}$

$U = \text{max}$

$U = \text{min}$

⑨ Power: Rate of doing work.

How fast work is done.

fast work $\Rightarrow$ High Power

Slow work $\Rightarrow$ Low Power

Average Power $\checkmark \langle \bar{P} \rangle_{\text{avg}} = \frac{\text{Total W/D}}{\text{time taken}}$

Instantaneous Power $\checkmark$

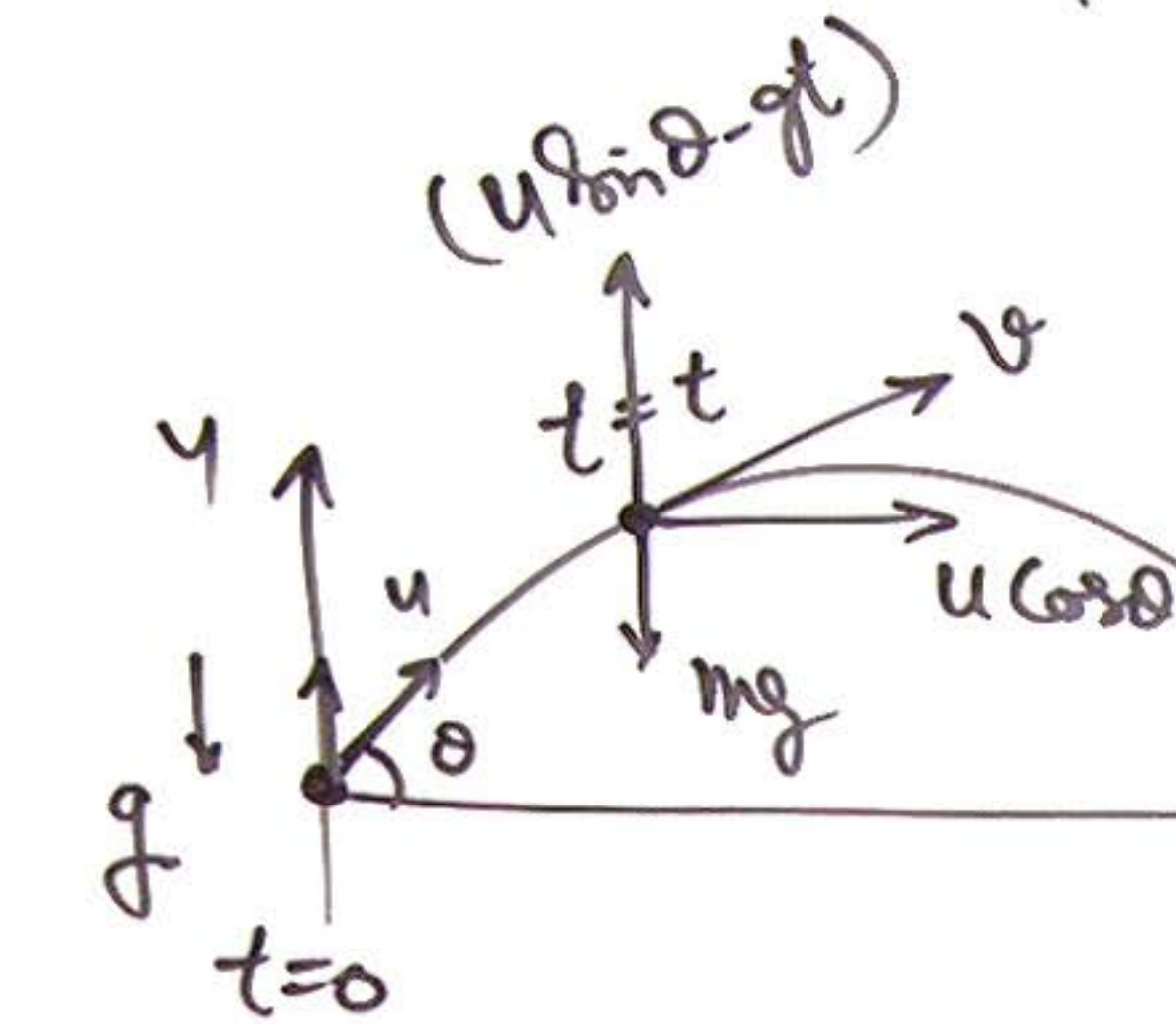
$$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$$

+ve $\rightarrow$ energy supply
-ve $\rightarrow$ energy gain

for any process.

+ve or -ve.

⑩ Calculation of Instantaneous power:



Q: power of gravity

Q: Power of friction $P_f =$



$$\begin{aligned}
 P_{\text{grav force}} &= \vec{F} \cdot \vec{v} \\
 &= mg \times (u \sin \theta - gt) \times \cos 180^\circ \\
 &= -mg(u \sin \theta - gt) \quad \text{Ans}
 \end{aligned}$$

$$P_f = -(\mu mg)(u - \mu g t)$$

Ans